

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 4



Mathematical Preparation

The set $2^{\mathbb{N}}$ is uncountable

Proof:

Suppose $2^{\mathbb{N}}$ is countably infinite.

Then we can enumerate all members of $2^{\mathbb{N}}$ as

$$2^{\mathbb{N}} = \{R_0, R_1, R_2, \dots\}$$

Consider the set $D = \{n \in \mathbb{N} : n \notin R_n\}$

Then $D \subseteq \mathbb{N}$ and so $D = R_k$ for some k

We ask the question whether $k \in R_k$

i. If **Yes**, then $k \notin D$ a contradiction as $D = R_k$

ii. If **No**, then $k \in D$ a contradiction as $D = R_k$ so $k \in R_k$

A contradiction

Mathematical Preparation

Closures

Intuitive Idea

Natural numbers N are **closed** under $+$

i.e. for given two natural numbers n, m we always have that $n + m \in N$

Natural numbers N are not closed under subtraction –

i.e there are two natural numbers n, m such that $n - m < N$,
for example $1, 2 \in N$ and $1 - 2 < N$

Integers Z are **closed** under–

Moreover Z is the **smallest set** containing N and closed under subtraction –

The set Z is called a **closure** of N under subtraction –

Mathematical Preparation

Consider the directed graph R
& the directed graph R^*

Observe that

$$R^* = R \cup \{(a_i, a_i) : i = 1, 2, 3, 4\} \\ \cup \{(a_2, a_4)\}$$

Observe

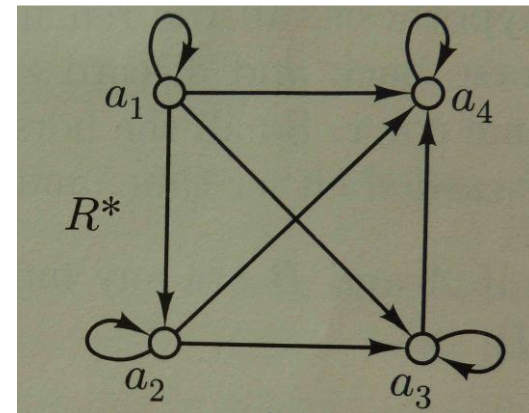
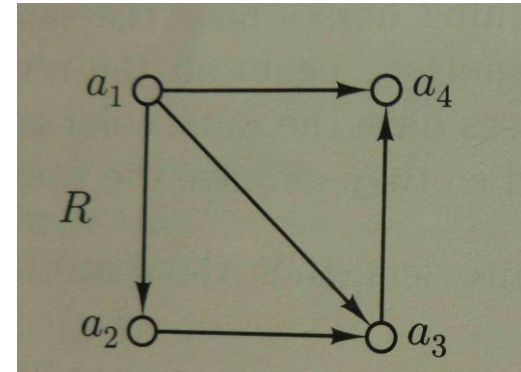
R is neither reflexive nor transitive

$$R \subseteq R^*$$

R^* is reflexive and transitive

Moreover R^* is also the **smallest set**
containing R that is **reflexive** and **transitive**

We call relation R^* the **reflexive, transitive closure** of R



Mathematical Preparation

Two Definitions of R^*

1. R^* is called a **reflexive, transitive closure** of R iff
 $R \subseteq R^*$ and R^* is reflexive and transitive and is the smallest set with these properties
2. Let R be a binary relation on a set A
The **reflexive, transitive closure** of R is the relation
 $R^* = \{(a, b) \in A \times A : \text{there is a path from } a \text{ to } b \text{ in } R\}$

Definition 2 is a much **simpler definition** and algorithmically more interesting as it uses a simple notion of a path

Mathematical Preparation

Definition 2 suggests the following algorithm for computing the reflexive transitive closure R^* of any given binary relation R over some finite set $A = \{a_1, a_2, \dots, a_n\}$

Algorithm 1

Initially $R^* := \phi$

for $i = 1, 2, \dots, n$ do

 for each i - tuple $(b_1, \dots, b_i) \in A^i$ do

 if $b_1, \dots, b_i$ is a **path** in R then add (b_1, b_i) to R^*

Running Time

$O(n^{n+1})$

Why?

In worst case n^n tuples to check.

Each tuple requires n operations

Mathematical Preparation

Algorithm 2

Initially $R^* := R \cup \{(a_i, a_i) : a_i \in A\}$

While there are three elements a_i, a_j, a_k

if $(a_i, a_j); (a_j, a_k) \in R^*$ but $(a_i, a_k) \notin R^*$

then add (a_i, a_k) to R^*

This algorithm is more natural & intuitive than the first.

Running Time

$O(n^5)$

Why?

In worst case n^2 pairs to be added - iterations

In each iteration all triples needs to be searched (**Why**) – $O(n^3)$

Languages over Σ

Definition

Any finite set is called an **alphabet**

Elements of the **alphabet** are called symbols of the alphabet

Alphabet is any finite set of **symbols**

Alphabet Notation

We use a symbol Σ to denote the **alphabet**

$\Sigma = \{0, 1\}$ is called a **binary alphabet**

Remember

Σ can be ϕ as empty set is a **finite set**

We want to study $\Sigma \neq \phi$

Languages over Σ

We call finite sequences of the alphabet Σ as **words/strings** over Σ

We denote by e/λ the **empty string** over Σ

Let Σ be an **alphabet**.

We denote by Σ^* the set of **strings** over Σ

Observe that $e \in \Sigma^*$

So $\Sigma^* \neq \emptyset$

Languages over Σ

Language Definition

Given an alphabet Σ , any set L such that $L \subseteq \Sigma^*$ is called a *language over Σ*

Fact1:

For any alphabet Σ , any language over Σ is *countable*

Fact2:

For any alphabet $\Sigma \neq \phi$, there are *uncountably many* languages over Σ

Concatenation

Informal Definition

Given an alphabet Σ and any words $x, y \in \Sigma^*$

We define informally a **concatenation** of words x, y as a word w obtained from x, y by writing the word x followed by the word y

We write the concatenation of words x, y as **$w = x \circ y$**

We use the symbol $\circ$ of concatenation when it is needed

Formally, otherwise we will write simply **$w = xy$**

Observe

$$w \circ e = e \circ w = w$$

$$(x \circ y) \circ z = x \circ (y \circ z) = x \circ y \circ z$$

Languages over Σ

Definition

A word $v \in \Sigma^*$ is a **substring** (sub-word) of w iff there are $x, y \in \Sigma^*$ such that

$$w = xvy$$

Remark: the words $x, y \in \Sigma^*$, i.e. they can also be empty

P1 w is a substring of w

P2 ϵ is a substring of any string (any word w)

as we have that $\epsilon w = w\epsilon = w$

Definition Let $w = xy$

x is called a **prefix** and y is called a **suffix** of w

Languages over Σ

Definition

We define a **power** w^i of w by Mathematical Induction as follows

$$\begin{aligned}w^0 &= e \\w^{i+1} &= w^i \circ w\end{aligned}$$

Reversal w^R of w is defined by induction over length $|w|$ of w as follows

1. If $|w| = 0$, then $w^R = w = e$
2. If $|w| = n + 1 > 0$, then $w = ua$ for some $a \in \Sigma$, and $u \in \Sigma^*$ and we define

$$w^R = au^R \text{ for } |u| < n + 1$$

For any $w, x \in \Sigma^*$

$$(wx)^R = x^R w^R$$

Languages over Σ

Languages are **sets** so we can define them in ways we did for sets, by **listing** elements (for small finite sets) or by giving a **property** $P(w)$ **defining** L , i.e. by setting

$$L = \{w \in \Sigma^* : P(w)\}$$

$$L_1 = \{w \in \{0, 1\}^* : w \text{ has an even number of } 0\text{'s} \}$$

$$L_2 = \{w \in \{a, b\}^* : w \text{ has } ab \text{ as a sub-string} \}$$

$$L_4 = \{e, 0, 1, 00, 01, 11, 10\}$$

Can we define L_4 using a property ?

Languages over Σ

Languages are **sets** so we can define set operations of union, intersection, generalized union, generalized intersection, complement, Cartesian product, ... etc ... of languages as we did for any sets

For example, given $L, L_1, L_2 \subseteq \Sigma^*$, we consider

$$L_1 \cup L_2, \quad L_1 \cap L_2, \quad L_1 - L_2, \\ -L = \Sigma^* - L, \quad L_1 \times L_2, \dots \text{ etc}$$

and we have that all properties of **algebra of sets** hold for any languages over a given alphabet Σ

Concatenation

Concatenation Definition

Given $L_1, L_2 \subseteq \Sigma^*$, a language

$$L_1 \circ L_2 = \{w \in \Sigma^* : w = xy \text{ for some } x \in L_1, y \in L_2\}$$

is called a **concatenation** of the languages L_1 and L_2

Concatenation can also be defined for languages on different domains

$$L_1 = \{w \in \{0, 1\}^* : w \text{ has an even number of 0's}\}$$

And $L_2 = \{w \in \{0, 1\}^* : w \text{ starts with 0 \& rest of the symbols (if present) are 1's}\}$

$$L_1 \circ L_2 = \{w \in \Sigma^* : w \text{ has an odd number of 0's}\}$$

Concatenation

Is Concatenation of languages commutative?

Observe that

$1000 \in L_1 \circ L_2$ **But** $1000 \notin L_2 \circ L_1$

This proves that

$$L_1 \circ L_2 \neq L_2 \circ L_1$$

Concatenation is **distributive** over **union** of languages

$$n \geq 2$$

$$(L_1 \cup L_2 \cup \dots \cup L_n) \circ L = (L_1 \circ L) \cup \dots \cup (L_n \circ L)$$

$$L \circ (L_1 \cup L_2 \cup \dots \cup L_n) = (L \circ L_1) \cup \dots \cup (L \circ L_n)$$